Excel Data Analytics Learning Packet

From Raw Data to Business Insights

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Welcome

This learning packet is designed to help you develop practical Excel skills by working with one real-world dataset from start to finish.

Instead of learning Excel functions and tools separately, you will use a single dataset to practice the same types of tasks a data or business analyst may encounter when working with real information: cleaning data, analyzing it, identifying patterns, building visualizations, and communicating findings.

The goal is not simply to memorize Excel formulas. The goal is to understand how Excel can be used to turn raw data into information that supports business decisions.

What You Will Learn

Throughout this packet, you will progressively work through five core areas:

01 — Data Cleaning

Learn how to prepare a raw dataset for analysis by identifying and correcting issues such as missing values, duplicates, inconsistent formatting, and incorrect data.

02 — Excel Formulas

Build a foundation in formulas and functions used to calculate, organize, compare, and analyze data.

03 — Pivot Tables & Analysis

Learn how to summarize large amounts of information and answer questions about trends, categories, performance, and relationships within the dataset.

04 — Dashboard Creation

Transform your analysis into an interactive and visually understandable dashboard using charts, key performance indicators, and other Excel tools.

05 — Business Reporting

Learn how to move beyond numbers by explaining what the data means, why the findings matter, and what actions a business or organization could consider.

How This Packet Works

You will work with one dataset throughout the entire packet.

Rather than completing isolated exercises with unrelated examples, each section builds on the previous one:

> **Raw Data → Clean Data → Analysis → Visualization → Business Insights**

You will first learn a concept, apply it to the dataset, and then use what you learned to complete a practical task.

By the end, you should have both a stronger understanding of Excel and a completed project that demonstrates your ability to use Excel for data analysis and business problem-solving.

Your Final Goal

By completing this packet, you should be able to take an unfamiliar dataset and confidently answer:

> **“How can I use Excel to turn this data into something useful?”**

The answer should involve more than creating a spreadsheet. You should be able to prepare the data, analyze it, visualize the results, and communicate what you discovered.

Let's get started.

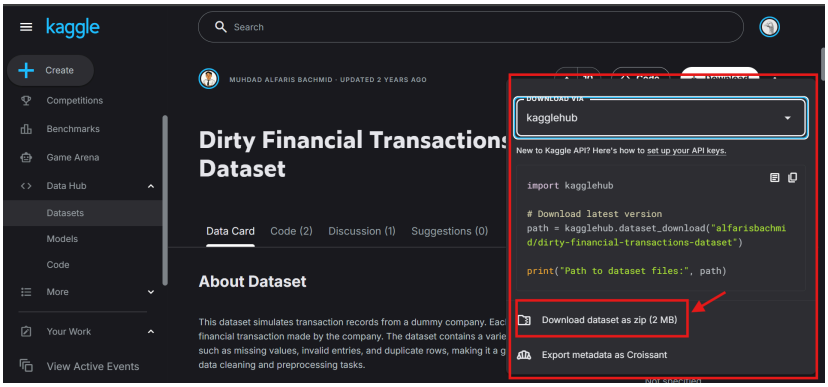
Phase 1 — Excel fundamentals

In this section, we will go over the necessary fundamentals to get you started with Excel. We'll be starting off with a blank worksheet before we download from Kaggle to start cleaning. You can use this as a starter project to display in your portfolio as proof as foundational knowledge in excel

Before we get started, we will be downloading a dataset from Kaggle. This is a follow along packet, so pick a dataset relevant to your desired industry.

The dataset we will be using in this worksheet in the **Dirty Financial Transactions Dataset** from Kaggle.

Link: https://www.kaggle.com/datasets/alfarisbachmid/dirty-financial-transactions-dataset



To download your selected dataset from Kaggle, click the download button at the top right of the screen. Then selected download dataset as a zip file.

Workbook/Sheet Organization

What is a Workbook?

An **Excel workbook** is the entire Excel file. A workbook can contain multiple worksheets, which are the individual tabs at the bottom of the file.

What is a Worksheet?

A **worksheet** is an individual sheet within a workbook where you enter and work with data.

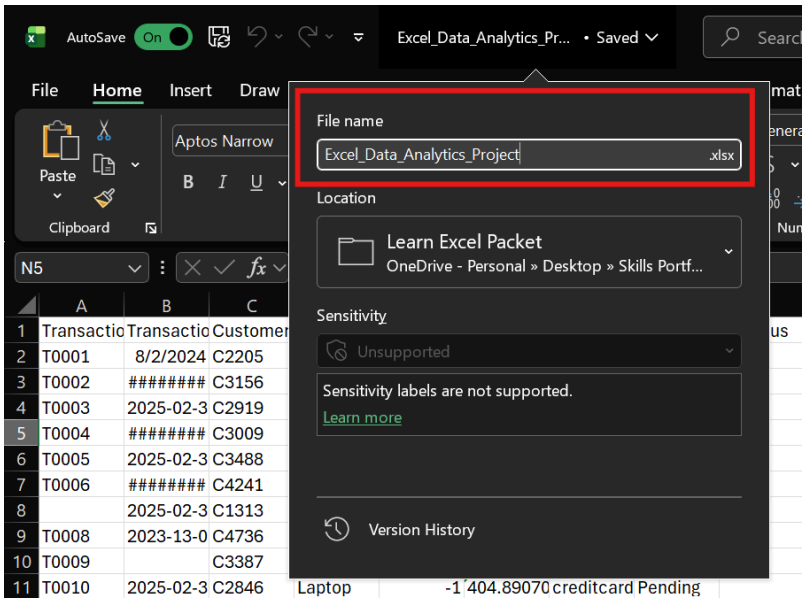
Step 1: Open a New Workbook

1. Open the Microsoft Excel Workbook.
2. Click the Folder name at the top of the taskbar next to the save button

3. Give the workbook a descriptive name related to your project.
4. Save the workbook.

Example:

`Excel_Data_Analytics_Project.xlsx`

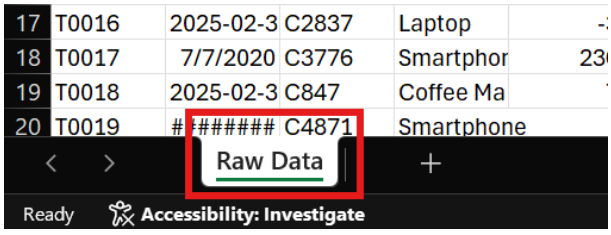


Step 1: Rename a Worksheet

1. Look at the worksheet tabs at the bottom of Excel.
2. Right-click **Sheet1**.
3. Select **Rename**.
4. Enter the name you want to use.
5. Press **Enter**.

For this project, rename your first sheet:

Raw Data

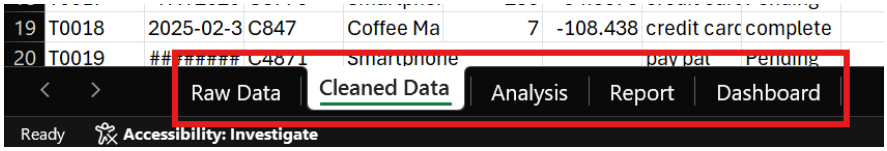


Step 1: Add Additional Worksheets

1. Click the **+** button next to the worksheet tabs.
2. A new worksheet will appear.
3. Rename the worksheet.
4. Repeat until you have the sheets you need.

For this project, create:

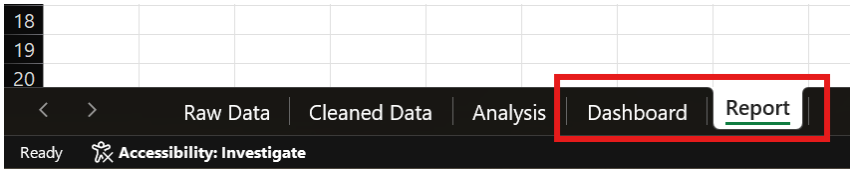
- **Raw Data**
- **Cleaned Data (Duplicate Raw Data, we'll use this in Phase 2)**
    - To duplicate a sheet in Microsoft Excel, the fastest way is to hold the Ctrl key (or Option key on a Mac) while clicking and dragging the sheet tab to a new spot along the bottom of your screen.
- **Analysis**
- **Dashboard**
- **Report**



Step 2: Reorder Worksheets

1. Click and hold the worksheet tab.
2. Drag it left or right.
3. Release it when it is in the desired position.

Organize your workbook so the sheets follow the order in which you will use them.



Practice

Create and organize your workbook using the five worksheets above.

Formatting

What is Formatting?

Formatting changes the appearance of your data to make it **easier to read, understand, and interpret**.

Formatting does not normally change the underlying value of the data.

Step 1: Format Column Headers

1. Select the row containing your column headers.
2. Select **Bold** from the Home tab.
3. Adjust the font size if necessary.
4. Use the alignment tools to position the headers appropriately.

Step 2: Adjust Column Width

1. Select the columns you want to resize.
2. Move your cursor between two column letters at the top.
3. Double-click the boundary between the columns.
4. Excel will automatically adjust the column width to fit the contents.

You can also click and drag the boundary to manually adjust the width.

Step 3: Format Numbers

1. Select the cells containing numerical values.
2. Go to the **Home** tab.
3. Locate the **Number** section.
4. Select the appropriate number format.

Common formats include:

- Number
- Currency
- Percentage
- Date
- Decimal

Number, Currency, Percentage, and Date Formatting

Excel formatting changes **how a value looks**, but it does not necessarily change what the value actually is.

For example:

1250

can be displayed as:

$1,250.00

when Excel recognizes it as a number.

However, if `1250` is stored as text, changing the cell to Currency may not work.

Steps

1. Select the column you want to format.
2. Look at the values in the cells.
3. Go to **Home** → **Number Format**.
4. Select the appropriate format:
    - Number
    - Currency
    - Percentage
    - Date
5. Check whether the values actually change.

If the formatting doesn't work:

Don't immediately keep trying different formats.

This may indicate that Excel is storing the values as **text**.

This is something you will learn to diagnose and correct in **Phase 2 — Data Cleaning**.

Step 4: Format Dates

1. Select the cells containing dates.
2. Go to **Home** → **Number Format**.
3. Select **Short Date** or **Long Date**.
4. Choose the format that makes the dates easiest to read.

Step 5: Apply Borders

1. Select the range of cells you want to format.
2. Go to **Home** → **Borders**.
3. Select the appropriate border option.

Avoid adding excessive borders. Use them to help separate or organize information.

Step 6: Apply Conditional Formatting

Conditional formatting allows Excel to automatically change the appearance of cells based on their values.

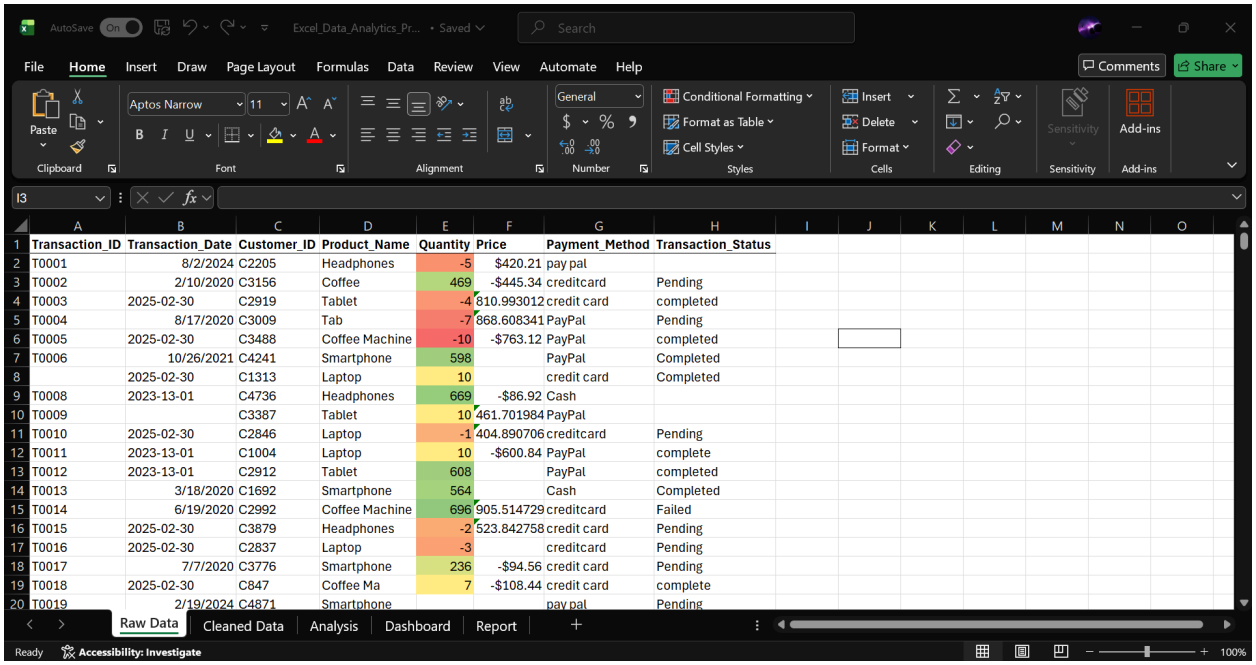
1. Select the data you want to analyze.
2. Go to **Home → Conditional Formatting**.
3. Choose a rule.
4. Select the condition you want to apply.
5. Choose the formatting.
6. Click **OK**.

Practice

Format your dataset so that:

- Headers are easy to identify.
- Columns are wide enough to read.
- Dates have a consistent format.
- Currency values display as currency.
- Percentages display as percentages.
- Numerical values use appropriate decimal places.

You're ending data set should look something like this:



Transaction_ID	Transaction_Date	Customer_ID	Product_Name	Quantity	Price	Payment_Method	Transaction_Status
T0001	8/2/2024	C2205	Headphones	-5	$420.21	pay pal	
T0002	2/10/2020	C3156	Coffee	469	-$445.34	creditcard	Pending
T0003	2025-02-30	C2919	Tablet	-4	810.993012	credit card	completed
T0004	8/17/2020	C3009	Tab	-7	868.608341	PayPal	Pending
T0005	2025-02-30	C3488	Coffee Machine	-10	-$763.12	PayPal	completed
T0006	10/26/2021	C4241	Smartphone	598		PayPal	Completed
	2025-02-30	C1313	Laptop	10		credit card	Completed
T0008	2023-13-01	C4736	Headphones	669	-$86.92	Cash	
T0009		C3387	Tablet	10	461.701984	PayPal	
T0010	2025-02-30	C2846	Laptop	-1	404.890706	creditcard	Pending
T0011	2023-13-01	C1004	Laptop	10	-$600.84	PayPal	complete
T0012	2023-13-01	C2912	Tablet	608		PayPal	completed
T0013	3/18/2020	C1692	Smartphone	564		Cash	Completed
T0014	6/19/2020	C2992	Coffee Machine	696	905.514729	creditcard	Failed
T0015	2025-02-30	C3879	Headphones	-2	523.842758	credit card	Pending
T0016	2025-02-30	C2837	Laptop	-3		creditcard	Pending
T0017	7/7/2020	C3776	Smartphone	236	-$94.56	credit card	Pending
T0018	2025-02-30	C847	Coffee Ma	7	-$108.44	credit card	complete
T0019	2/19/2024	C4871	Smartphone			pay pal	Pending

Tables

What is an Excel Table?

An Excel Table converts a regular range of data into a structured dataset that Excel can easily sort, filter, expand, and reference.

Step 1: Select Your Dataset

1. Go to your **Raw Data** worksheet.
2. Click inside your dataset.
3. Press **Ctrl + A** to select the dataset.

Step 2: Create the Table

1. Go to **Insert**.
2. Select **Table**.
3. Excel will display the selected range.
4. Make sure **My table has headers** is checked.
5. Click **OK**.

Step 3: Name Your Table

1. Click anywhere inside the table.
2. Select the **Table Design** tab.
3. Find the **Table Name** box.
4. Enter a descriptive name.

For example:

```
RawData
```

Avoid using spaces in table names.

Step 4: Explore the Table

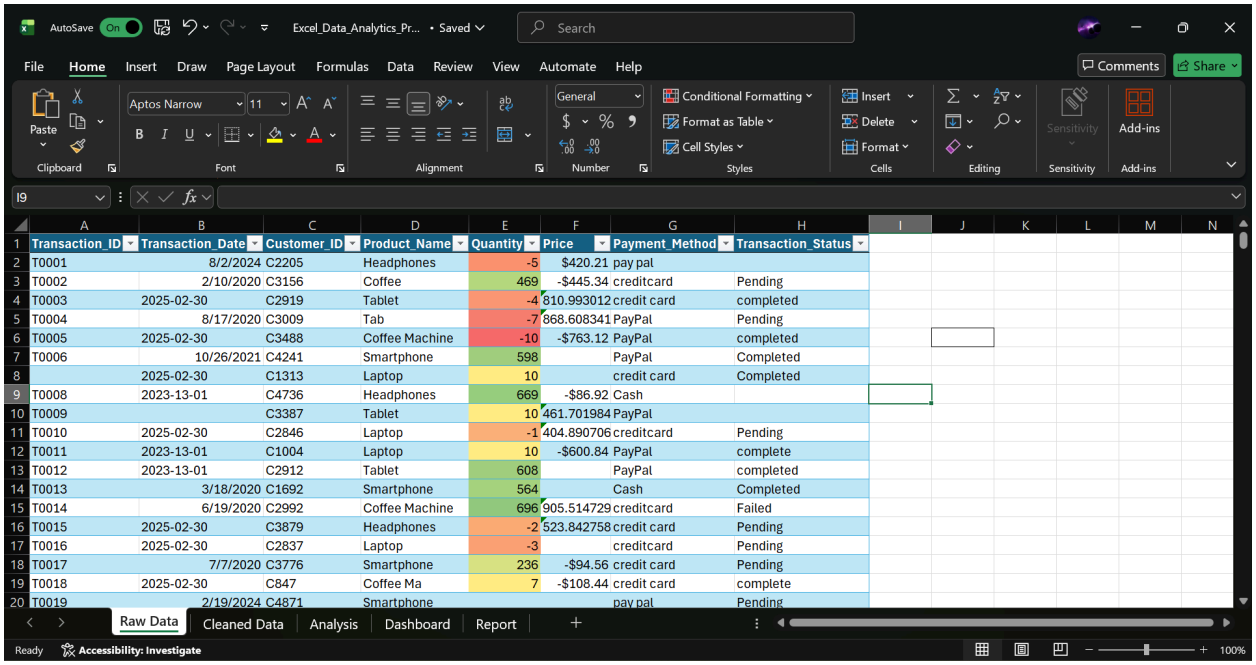
Notice that Excel automatically adds:

- Filter buttons to the headers
- Table formatting
- Automatic expansion when new rows are added

Practice

Convert your dataset into an Excel Table and give it a descriptive name.

You’re ending result should look something like this:



	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	Transaction_ID	Transaction_Date	Customer_ID	Product_Name	Quantity	Price	Payment_Method	Transaction_Status						
2	T0001	8/2/2024	C2205	Headphones	-5	$420.21	pay pal							
3	T0002	2/10/2020	C3156	Coffee	469	-$445.34	creditcard	Pending						
4	T0003	2025-02-30	C2919	Tablet	-4	810.993012	credit card	completed						
5	T0004	8/17/2020	C3009	Tab	-7	868.608341	PayPal	Pending						
6	T0005	2025-02-30	C3488	Coffee Machine	-10	-$763.12	PayPal	completed						
7	T0006	10/26/2021	C4241	Smartphone	598		PayPal	Completed						
8		2025-02-30	C1313	Laptop	10		credit card	Completed						
9	T0008	2023-13-01	C4736	Headphones	669	-$86.92	Cash							
10	T0009		C3387	Tablet	10	461.701984	PayPal							
11	T0010	2025-02-30	C2846	Laptop	-1	404.890706	creditcard	Pending						
12	T0011	2023-13-01	C1004	Laptop	10	-$600.84	PayPal	complete						
13	T0012	2023-13-01	C2912	Tablet	608		PayPal	completed						
14	T0013	3/18/2020	C1692	Smartphone	564		Cash	Completed						
15	T0014	6/19/2020	C2992	Coffee Machine	696	905.514729	creditcard	Failed						
16	T0015	2025-02-30	C3879	Headphones	-2	523.842758	credit card	Pending						
17	T0016	2025-02-30	C2837	Laptop	-3		creditcard	Pending						
18	T0017	7/7/2020	C3776	Smartphone	236	-$94.56	credit card	Pending						
19	T0018	2025-02-30	C847	Coffee Ma	7	-$108.44	credit card	complete						
20	T0019	2/19/2024	C4871	Smartphone			pay pal	Pending						

Sorting/Filtering

Sorting

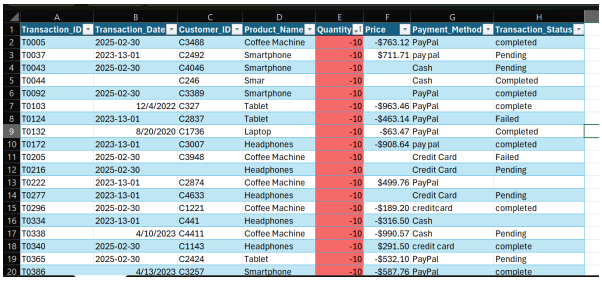
Sorting changes the order of your data based on a selected column.

Step 1: Sort Data

1. Click the filter arrow on the column you want to sort.
2. Choose **Sort A to Z** or **Sort Z to A** for text.
3. For numbers, choose **Sort Smallest to Largest** or **Sort Largest to Smallest**.
4. Review the results.

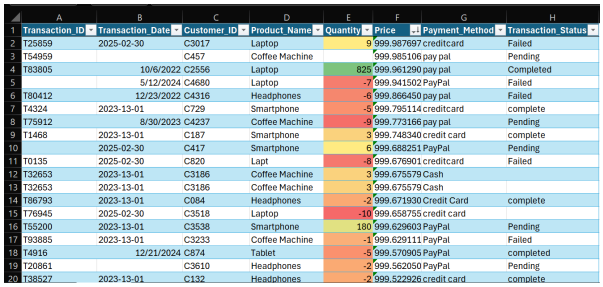
These are what some of your results will look like:

Quantity Smallest to Largest



	A	B	C	D	E	F	G	H
1	Transaction_ID	Transaction_Date	Customer_ID	Product_Name	Quantity	Price	Payment_Method	Transaction_Status
2	T0005	2025-02-30	C3488	Coffee Machine	-10	-$763.12	PayPal	completed
3	T0037	2023-13-01	C2492	Smartphone	-10	$711.71	pay pal	Pending
4	T0043	2025-02-30	C4046	Smartphone	-10		Cash	Pending
5	T0044		C246	Smar	-10		Cash	Completed
6	T0092	2025-02-30	C3389	Smartphone	-10		PayPal	completed
7	T0103	12/4/2022	C327	Tablet	-10	-$963.46	PayPal	complete
8	T0124	2023-13-01	C2837	Tablet	-10	-$463.14	PayPal	Failed
9	T0132	8/20/2020	C1736	Laptop	-10	-$63.47	PayPal	Completed
10	T0172	2023-13-01	C3007	Headphones	-10	-$908.64	pay pal	completed
11	T0205	2025-02-30	C3948	Coffee Machine	-10		Credit Card	Failed
12	T0216	2025-02-30		Headphones	-10		Credit Card	Pending
13	T0222	2023-13-01	C2874	Coffee Machine	-10	$499.76	PayPal	
14	T0277	2023-13-01	C4633	Headphones	-10		Credit Card	Pending
15	T0296	2025-02-30	C1221	Coffee Machine	-10	-$189.20	creditcard	completed
16	T0334	2023-13-01	C441	Headphones	-10	-$316.50	Cash	
17	T0338	4/10/2023	C4411	Coffee Machine	-10	-$990.57	Cash	Pending
18	T0340	2025-02-30	C1143	Headphones	-10	$291.50	credit card	complete
19	T0365	2025-02-30	C2424	Tablet	-10	-$532.10	PayPal	Pending
20	T0386	4/13/2023	C3257	Smartphone	-10	-$587.76	PayPal	complete

Price Largest to Smallest



	A	B	C	D	E	F	G	H
1	Transaction_ID	Transaction_Date	Customer_ID	Product_Name	Quantity	Price	Payment_Method	Transaction_Status
2	T25859	2025-02-30	C3017	Laptop	9	999.987697	creditcard	Failed
3	T54959		C457	Coffee Machine		999.985106	pay pal	Pending
4	T83805	10/6/2022	C2556	Laptop	825	999.961290	pay pal	Completed
5		5/12/2024	C4680	Laptop	-7	999.941502	PayPal	Failed
6	T80412	12/23/2022	C4316	Headphones	-6	999.866450	pay pal	Failed
7	T4324	2023-13-01	C729	Smartphone	-5	999.795114	creditcard	complete
8	T75912	8/30/2023	C4237	Coffee Machine	-9	999.773166	pay pal	Pending
9	T1468	2023-13-01	C187	Smartphone	3	999.748340	credit card	complete
10		2025-02-30	C417	Smartphone	6	999.688251	PayPal	Pending
11	T0135	2025-02-30	C820	Lapt	-8	999.676901	creditcard	Failed
12	T32653	2023-13-01	C3186	Coffee Machine	3	999.675579	Cash	
13	T32653	2023-13-01	C3186	Coffee Machine	3	999.675579	Cash	
14	T86793	2023-13-01	C084	Headphones	-2	999.671930	Credit Card	complete
15	T76945	2025-02-30	C3518	Laptop	-10	999.658755	credit card	
16	T55200	2023-13-01	C3538	Smartphone	180	999.629603	PayPal	Pending
17	T93885	2023-13-01	C3233	Coffee Machine	-1	999.629111	PayPal	Failed
18	T4916	12/21/2024	C874	Tablet	-5	999.570905	PayPal	completed
19	T20861		C3610	Headphones	-2	999.562050	PayPal	Pending
20	T38527	2023-13-01	C132	Headphones	-2	999.522926	credit card	complete

For Now we are just gonna sort smallest to largest for the Transaction_ID column

Important

Make sure you are sorting the **entire table**, not just one column. Otherwise, your data could become mismatched.

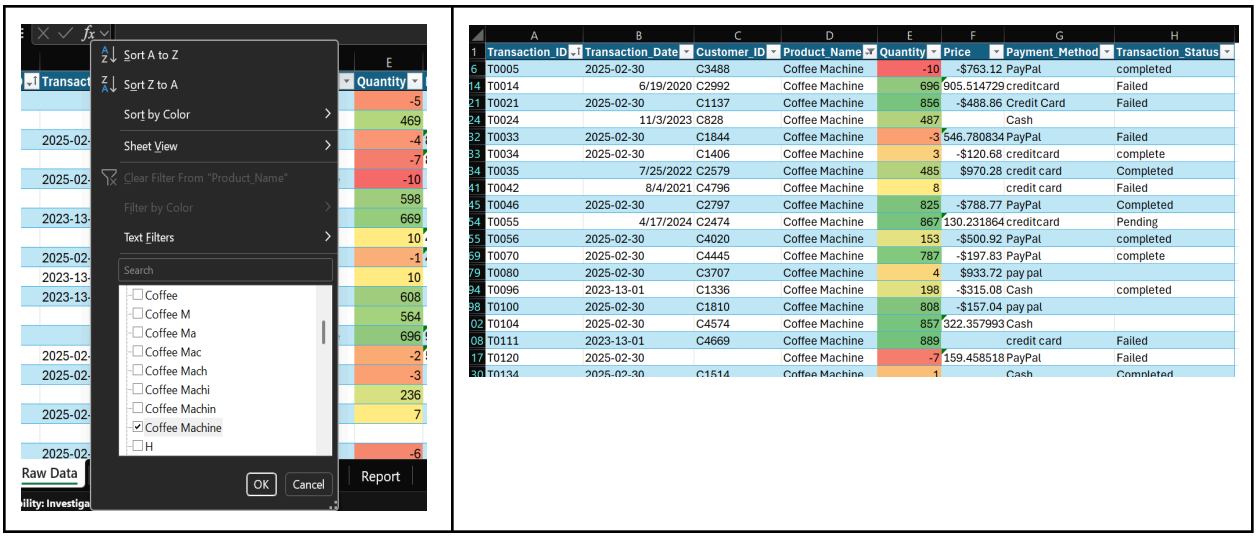
Filtering

Filtering temporarily hides rows that don't meet your selected criteria.

Step 1: Apply a Filter

1. Click the filter arrow on the column you want to filter.
2. Uncheck **Select All**.
3. Select the value or values you want to see.
4. Click **OK**.

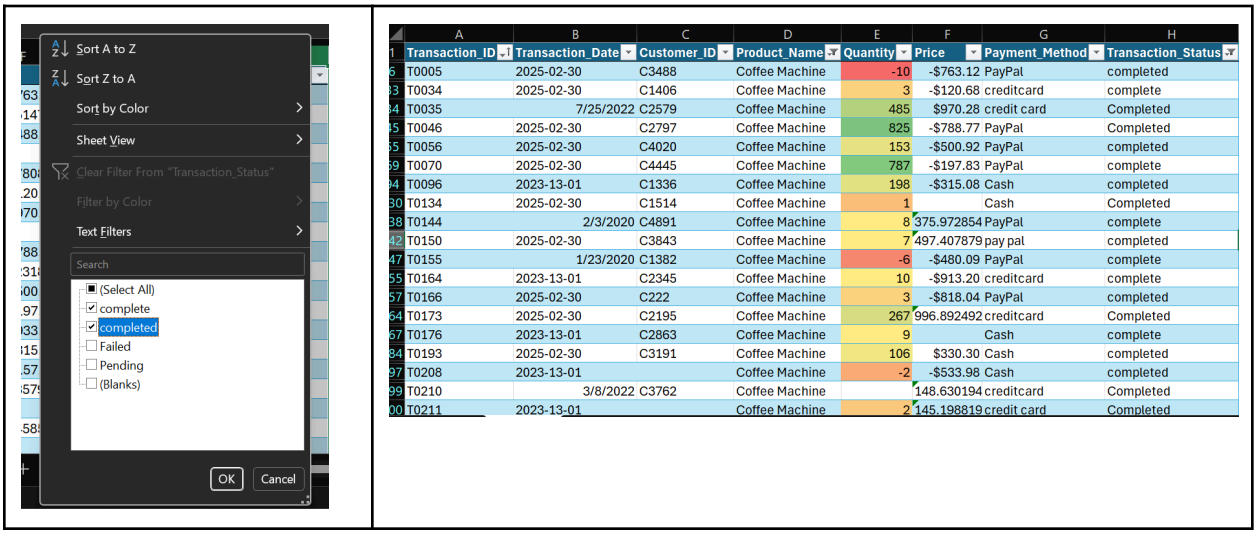
Excel will display only the rows that meet your criteria.



Transaction_ID	Transaction_Date	Customer_ID	Product_Name	Quantity	Price	Payment_Method	Transaction_Status
T0005	2025-02-30	C3488	Coffee Machine	-10	-$763.12	PayPal	completed
T0014	6/19/2020	C2992	Coffee Machine	696	905.514729	creditcard	Failed
T0021	2025-02-30	C1137	Coffee Machine	856	-$488.86	Credit Card	Failed
T0024	11/3/2023	C828	Coffee Machine	487		Cash	
T0033	2025-02-30	C1844	Coffee Machine	-3	546.780834	PayPal	Failed
T0034	2025-02-30	C1406	Coffee Machine	3	-$120.68	creditcard	complete
T0035	7/25/2022	C2579	Coffee Machine	485	$970.28	credit card	Completed
T0042	8/4/2021	C4796	Coffee Machine	8		credit card	Failed
T0046	2025-02-30	C2797	Coffee Machine	825	-$788.77	PayPal	Completed
T0055	4/17/2024	C2474	Coffee Machine	867	130.231864	creditcard	Pending
T0056	2025-02-30	C4020	Coffee Machine	153	-$500.92	PayPal	completed
T0070	2025-02-30	C4445	Coffee Machine	787	-$197.83	PayPal	complete
T0080	2025-02-30	C3707	Coffee Machine	4	$933.72	pay pal	
T0096	2023-13-01	C1336	Coffee Machine	198	-$315.08	Cash	completed
T0100	2025-02-30	C1810	Coffee Machine	808	-$157.04	pay pal	
T0104	2025-02-30	C4574	Coffee Machine	857	322.357993	Cash	
T0111	2023-13-01	C4669	Coffee Machine	889		credit card	Failed
T0120	2025-02-30		Coffee Machine	-7	159.458518	PayPal	Failed
T0134	2025-02-30	C1514	Coffee Machine	1		Cash	Completed

Step 2: Filter Using Multiple Conditions

1. Open the filter menu for the first column.
2. Select your desired value.
3. Open another column's filter menu.
4. Select another value.
5. Excel will display records that meet both conditions.

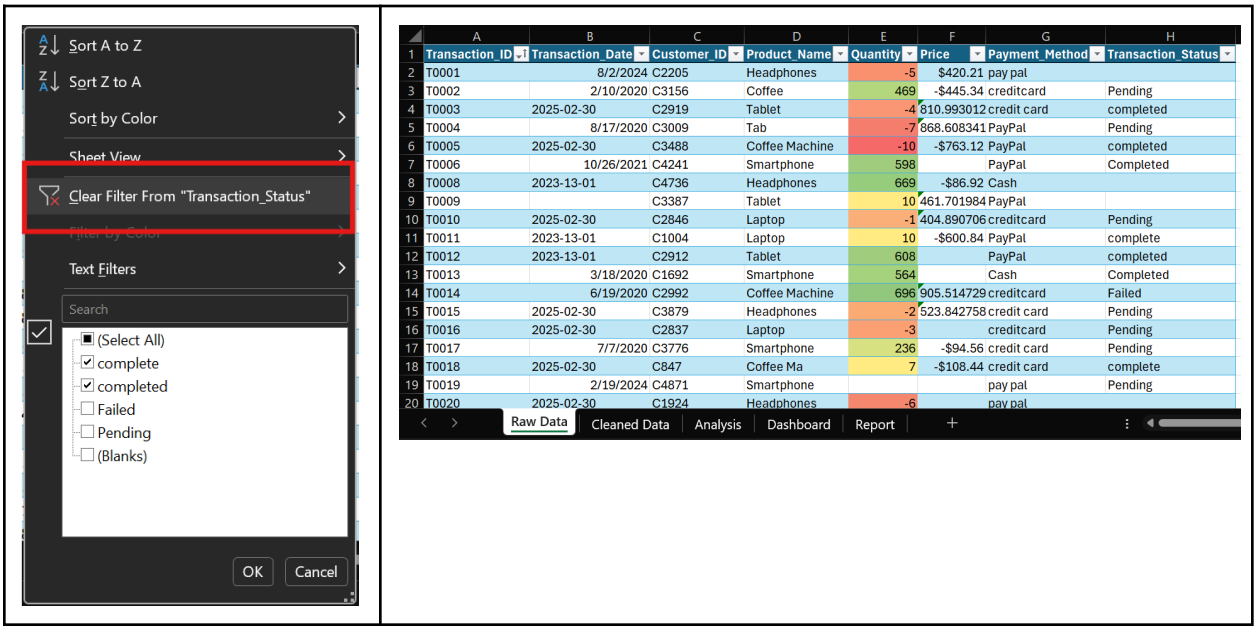


Transaction_ID	Transaction_Date	Customer_ID	Product_Name	Quantity	Price	Payment_Method	Transaction_Status
T0005	2025-02-30	C3488	Coffee Machine	-10	-$763.12	PayPal	completed
T0034	2025-02-30	C1406	Coffee Machine	3	-$120.68	creditcard	complete
T0035	7/25/2022	C2579	Coffee Machine	485	$970.28	credit card	Completed
T0046	2025-02-30	C2797	Coffee Machine	825	-$788.77	PayPal	Completed
T0056	2025-02-30	C4020	Coffee Machine	153	-$500.92	PayPal	completed
T0070	2025-02-30	C4445	Coffee Machine	787	-$197.83	PayPal	complete
T0096	2023-13-01	C1336	Coffee Machine	198	-$315.08	Cash	completed
T0134	2025-02-30	C1514	Coffee Machine	1		Cash	Completed
T0144	2/3/2020	C4891	Coffee Machine	8	375.972854	PayPal	complete
T0150	2025-02-30	C3843	Coffee Machine	7	497.407879	pay pal	completed
T0155	1/23/2020	C1382	Coffee Machine	-6	-$480.09	PayPal	complete
T0164	2023-13-01	C2345	Coffee Machine	10	-$913.20	creditcard	complete
T0166	2025-02-30	C222	Coffee Machine	3	-$818.04	PayPal	completed
T0173	2025-02-30	C2195	Coffee Machine	267	996.892492	creditcard	Completed
T0176	2023-13-01	C2863	Coffee Machine	9		Cash	complete
T0193	2025-02-30	C3191	Coffee Machine	106	$330.30	Cash	completed
T0208	2023-13-01		Coffee Machine	-2	-$533.98	Cash	completed
T0210	3/8/2022	C3762	Coffee Machine		148.630194	creditcard	Completed
T0211	2023-13-01		Coffee Machine	2	145.198819	credit card	Completed

Step 3: Clear a Filter

1. Click the filter arrow.
2. Select **Clear Filter From...**

Your original data will become visible again.



	A	B	C	D	E	F	G	H
1	Transaction_ID	Transaction_Date	Customer_ID	Product_Name	Quantity	Price	Payment_Method	Transaction_Status
2	T0001	8/2/2024	C2205	Headphones	-5	$420.21	pay pal	
3	T0002	2/10/2020	C3156	Coffee	469	-$445.34	creditcard	Pending
4	T0003	2025-02-30	C2919	Tablet	-4	810.993012	credit card	completed
5	T0004	8/17/2020	C3009	Tab	-7	868.608341	PayPal	Pending
6	T0005	2025-02-30	C3488	Coffee Machine	-10	-$763.12	PayPal	completed
7	T0006	10/26/2021	C4241	Smartphone	598		PayPal	Completed
8	T0008	2023-13-01	C4736	Headphones	669	-$86.92	Cash	
9	T0009		C3387	Tablet	10	461.701984	PayPal	
10	T0010	2025-02-30	C2846	Laptop	-1	404.890706	creditcard	Pending
11	T0011	2023-13-01	C1004	Laptop	10	-$600.84	PayPal	complete
12	T0012	2023-13-01	C2912	Tablet	608		PayPal	completed
13	T0013	3/18/2020	C1692	Smartphone	564		Cash	Completed
14	T0014	6/19/2020	C2992	Coffee Machine	696	905.514729	creditcard	Failed
15	T0015	2025-02-30	C3879	Headphones	-2	523.842758	credit card	Pending
16	T0016	2025-02-30	C2837	Laptop	-3		creditcard	Pending
17	T0017	7/7/2020	C3776	Smartphone	236	-$94.56	credit card	Pending
18	T0018	2025-02-30	C847	Coffee Ma	7	-$108.44	credit card	complete
19	T0019	2/19/2024	C4871	Smartphone			pay pal	Pending
20	T0020	2025-02-30	C1924	Headphones	-6		pay pal	

Practice

Using your dataset:

1. Sort one numerical column from largest to smallest.
2. Sort one text column alphabetically.
3. Filter the dataset to show one specific category.
4. Apply filters to two different columns at the same time.
5. Clear all filters.

Relative vs. Absolute References

Cell references tell Excel **which cells to use in a calculation**.

There are two important types of references:

Relative Reference

A relative reference changes when you copy a formula to another cell.

Example:

=A2*B2

If you copy this formula down one row, Excel changes it to:

=A3*B3

Absolute Reference

An absolute reference stays fixed when you copy a formula.

It uses `$`.

Example:

=A2*F1

If you copy this formula down, it becomes:

=A3*F1

The `A2` changes, but `$F$1` stays the same.

Helpful Shortcut

When editing a cell reference in a formula, press **F4** to cycle through different reference types.

Basic Formulas

Formulas allow Excel to perform calculations automatically.

Every Excel formula begins with:

=

Basic Arithmetic

Excel can perform basic mathematical operations.

Addition

=A2+B2

Subtraction

=A2-B2

Multiplication

=A2*B2

Division

=A2/B2

Common Excel Functions

SUM

Adds numbers together.

=SUM(A2:A20)

Steps

1. Select an empty cell.
2. Type `=SUM(`
3. Select the range you want to add.
4. Type `)`.
5. Press **Enter**.

AVERAGE

Calculates the average of a group of numbers.

=AVERAGE(A2:A20)

Steps

1. Select an empty cell.
2. Type `=AVERAGE(`

3. Select your numerical range.
4. Type).
5. Press **Enter**.

MIN

Finds the smallest value.

=MIN(A2:A20)

MAX

Finds the largest value.

=MAX(A2:A20)

COUNT

Counts cells containing numbers.

=COUNT(A2:A20)

COUNTA

Counts cells that aren't empty.

=COUNTA(A2:A20)

Phase 2 — Data cleaning

Assessing, Cleaning, and Preparing Your Data

Raw datasets are not always ready for analysis. They may contain missing values, duplicate records, inconsistent formatting, incorrect data types, or information that is unnecessary for your analysis.

In this phase, you will first **inspect and assess your dataset** to identify problems. You will then use Excel to correct the issues you actually find.

> **Important:** Do not assume that your dataset has every type of cleaning problem listed in this phase. The purpose of the assessment is to determine which issues are actually present.

2.1 Assess Your Dataset Before Cleaning

Before changing anything, take some time to understand the dataset you are working with.

Your dataset may come from a source such as Kaggle. If the dataset's description, documentation, or data dictionary identifies known issues, use that information as a starting point. However, **do not rely only on the dataset description**. You should also inspect the data yourself.

Step 1 — Review the Dataset Documentation

If your dataset came from Kaggle:

1. Open the dataset's Kaggle page.
2. Read the **About Dataset** or description.
3. Look for information about:
    - Missing values
    - Duplicate records
    - Data definitions
    - Column descriptions
    - Known errors
    - Data collection methods
    - Formatting or data-quality issues
4. Take note of any problems the dataset creator identifies.

financial transaction made by the company. The dataset contains a variety of data quality issues, such as missing values, invalid entries, and duplicate rows, making it a good candidate for practicing data cleaning and preprocessing tasks.

Columns:

1. Transaction_ID (String)
   A unique identifier for each transaction (e.g., T0001, T0002, T0003). Some values are missing, and others may contain special characters or incorrect formatting.
2. Transaction_Date (Date)
   The date of the transaction in YYYY-MM-DD format. There are some invalid dates (e.g., 2023-13-01, 2025-02-30) and missing values.
3. Customer_ID (String)
   The identifier for the customer (e.g., C001, C002, C003). There are missing values and duplicated IDs.
4. Product_Name (String)
   The name of the product bought (e.g., "Laptop", "Smartphone", "Coffee Machine"). Some product names are misspelled, and others are missing or have extra spaces.
5. Quantity (Integer)
   The number of items purchased in the transaction. Some rows contain negative values or outlier quantities (e.g., 1000 items for a single product).
6. Price (Float)
   The price of the product in USD. Some prices are negative, missing, or contain unexpected symbols (e.g., "$300", "price").
7. Payment_Method (Categorical: String)
   The method of payment used for the transaction (e.g., "Credit Card", "Cash", "PayPal"). Some entries are misspelled or not in the expected categories.
8. Transaction_Status (Categorical: String)
   The status of the transaction (e.g., "Completed", "Pending", "Failed"). Some values are inconsistent, such as using "complete" instead of "Completed", or having null values.

Key Features:
• Missing values: There are missing values in columns like Transaction_Date, Customer_ID, and Transaction_Status.
• Invalid/Incorrect Data: There are invalid dates (e.g., 2023-13-01, 2025-02-30) and negative values for Quantity and Price. Some rows have prices with symbols or letters like "$300" and "price".
• Duplicated data: Some customer IDs and transaction IDs might be duplicated.
• Inconsistent data: Payment method names like "Credit card" are inconsistent (case and spacing issues), as are Transaction_Status values like "completed" instead of "Completed".
• Outliers: The dataset contains extreme values like a quantity of 1000 items in a single transaction.

Step 2 — Inspect the Data Yourself

Go to your **Cleaned Data** sheet and review the columns and records.

Look for:

- Blank cells
- Duplicate records
- Numbers stored as text
- Inconsistent dates
- Different spellings of the same category
- Inconsistent capitalization
- Extra spaces
- Columns containing multiple pieces of information
- Columns that aren't useful for your analysis
- Values that appear unusual or incorrect

Don't immediately change anything.

Your goal at this stage is to **identify the problems first**.

Step 3 — Create a Dataset Quality Assessment

Before cleaning the data, record what you found.

Create a small table on your **Cleaned Data** sheet or in your learning packet:

Cleaning Issue	Found?	Location/Column	Action Needed
Missing values	Yes/No	_____	_____
Duplicates	Yes/No	_____	_____
Incorrect data types	Yes/No	_____	_____
Inconsistent spelling	Yes/No	_____	_____

Inconsistent capitalization	Yes/No	_____	_____
Extra spaces	Yes/No	_____	_____
Inconsistent date formatting	Yes/No	_____	_____
Columns to split/combine	Yes/No	_____	_____
Unnecessary columns	Yes/No	_____	_____
Data validation needed	Yes/No	_____	_____

Example

Suppose you inspect your dataset and discover:

- 12 missing values in the **Age** column
- 3 duplicate records
- Dates are formatted inconsistently
- No major spelling problems
- No columns need to be combined
- One column isn't needed for your analysis

Your assessment could look like:

Cleaning Issue	Found?	Location/Column	Action Needed
Missing values	Yes	Age	Investigate and handle

Duplicates	Yes	Entire dataset	Remove duplicates
Incorrect data types	No	—	None
Inconsistent spelling	No	—	None
Inconsistent capitalization	No	—	None
Extra spaces	No	—	None
Inconsistent date formatting	Yes	Date	Standardize
Columns to split/combine	No	—	None
Unnecessary columns	Yes	Column X	Remove
Data validation needed	Yes	Category	Create dropdown

This assessment becomes your **cleaning plan**.

For example:

Issue	**Found?**	**Column(s)**	**What will I do?**

Missing values	Yes	Transaction_ID, Date, Customer_ID, Status	Investigate; fill only when appropriate (mark out with black to flag issue)
Invalid dates	Yes	Transaction_Date	Convert valid dates; flag impossible dates for history and reporting
Duplicate transactions	Yes	Transaction_ID	Investigate before removing
Duplicate customer IDs	Yes	Customer_ID	**Do not automatically remove**
Misspelled products	Yes	Product_Name	Standardize using known product list
Extra spaces	Yes	Product_Name, Payment_Method, Status	Remove with TRIM, or standardize
Negative quantity	Yes	Quantity	Investigate; likely invalid
Extreme quantity	Yes	Quantity	Investigate as potential outlier
Invalid prices	Yes	Price	Convert valid entries; flag/remove invalid text
Inconsistent payment methods	Yes	Payment_Method	Standardize

Inconsistent statuses	Yes	Transaction_Status	Standardize
Missing status	Yes	Transaction_Status	Determine whether it can be reasonably filled

---

2.2 Create Your Cleaned Data Sheet

Before making any changes, **keep your original dataset untouched**.

Steps

1. Go to the **Raw Data** sheet.
2. Select the entire dataset.
3. Press `Ctrl + C`.
4. Go to the **Cleaned Data** sheet.
5. Click cell **A1**.
6. Press `Ctrl + V`.
7. Confirm that the copied dataset matches the Raw Data sheet.

From this point forward, perform your cleaning on **Cleaned Data**.

> **Important:** Your Raw Data sheet should remain unchanged. This gives you an original copy that you can return to if something goes wrong.

2.3 Clean the Issues You Identified

Now return to your Dataset Quality Assessment.

Only perform the cleaning steps that apply to your dataset.

For example, if you found missing values, complete the Missing Values section. If you did not find duplicates, you don't need to artificially create or remove any.

Missing Values

Steps

1. Identify the column containing missing values.
2. Click the filter arrow for that column.
3. Select **(Blanks)**.
4. Review the records containing missing information.
5. Determine the appropriate action.

Depending on the dataset, you may:

- Leave the value blank.
- Replace it with an appropriate value.
- Calculate the value if it can reasonably be determined.
- Remove the record if the missing information makes it unusable.
- Flag with conditional formatting

> **Do not automatically replace blank cells with 0.** A blank value and zero can represent different things.

Duplicates

Steps

1. Select your dataset.
2. Go to **Data** → **Remove Duplicates**.
3. Make sure **My data has headers** is selected.
4. Select the columns Excel should use to identify duplicates.
5. Click **OK**.
6. Review Excel's results.

> If your dataset has a unique ID, such as an Order ID or Customer ID, use it when appropriate.

Incorrect Data Types

Check whether each column contains the type of information it is supposed to contain.

For example:

Column	Expected Type
Customer Name	Text
Sales	Number/Currency
Date	Date
Discount	Percentage

Steps

1. Review each column.
2. Identify values that don't match the expected data type.
3. Select the affected cells or column.
4. Use **Home → Number Format** to apply the appropriate format.
5. If Excel identifies a number stored as text, use **Convert to Number** when appropriate.

Inconsistent Spelling & Capitalization

Look for situations such as:

Atlanta

atlanta

ATLANTA

These may represent the same category but can be treated differently during analysis.

Steps

1. Identify inconsistent values.
2. Use `Ctrl + H` to open **Find & Replace**.
3. Enter the incorrect value under **Find what**.
4. Enter the standardized value under **Replace with**.
5. Select **Replace All**.
6. Review the results.

You can also use:

=UPPER(A2)

=LOWER(A2)

=PROPER(A2)

depending on how you want the text standardized.

Extra Spaces

Extra spaces can cause problems when filtering, matching, or analyzing text.

For example:

" Atlanta"

"Atlanta "

" Atlanta "

Use the **TRIM** function:

=TRIM(A2)

Steps

1. Create a temporary column.
2. Enter the TRIM formula.
3. Copy the formula down.
4. Review the cleaned results.
5. Copy the results.

6. Use **Paste Special** → **Values** if you need to replace the original values.
7. Remove the temporary column.

Date Formatting

If your dataset contains dates in different formats, standardize them.

Examples:

1/5/2026

01/05/2026

January 5, 2026

Steps

1. Select the date column.
2. Go to **Home** → **Number Format**.
3. Select the desired date format.
4. If necessary, select **More Number Formats** → **Date**.
5. Choose your preferred format.
6. Click **OK**.

> Remember: changing the appearance of a date is different from converting text into an actual Excel date.

Splitting or Combining Columns

Only use this step if your assessment shows that a column needs to be split or combined.

To Split

Use:

Data → **Text to Columns**

For example:

John Smith

can become:

First Name	Last Name
John	Smith

To Combine

Use:

=A2&" "&B2

For example:

John + Smith

becomes:

John Smith

Removing Unnecessary Columns

Only remove columns that you have determined are unnecessary for your analysis.

Steps

1. Review the column.
2. Ask:

   Will this information be needed for my analysis, calculations, or final report?

3. If not, select the column.
4. Right-click the column header.
5. Select **Delete**.
6. Review the remaining dataset.

   Do not delete a column simply because you don't understand it. Check its purpose first.

Data Validation

Data validation helps prevent incorrect or inconsistent information from being entered into a dataset.

For example, if a **Region** column should only contain:

- North
- South
- East
- West

you can create a dropdown list.

Steps

1. Select the cells where the rule should apply.
2. Go to **Data** → **Data Validation**.
3. Under **Allow**, select **List**.
4. Enter the allowed values:

North,South,East,West

5. Make sure **In-cell dropdown** is selected.
6. Click **OK**.
7. Test the dropdown.

2.4 Final Data Quality Check

Once you've completed the necessary cleaning steps, go back to your original assessment.

Review each issue and confirm that it has been addressed.

Cleaning Issue	Initially Found?	Cleaned ?
Missing values	✓	✓

Duplicates	✓	✓
Incorrect data types	—	N/A
Inconsistent spelling	—	N/A
Extra spaces	✓	✓
Date formatting	✓	✓
Split/combine columns	—	N/A
Unnecessary columns	✓	✓
Data validation	✓	✓

N/A is important here. It means the issue wasn't present or wasn't applicable to your dataset.

Final Questions

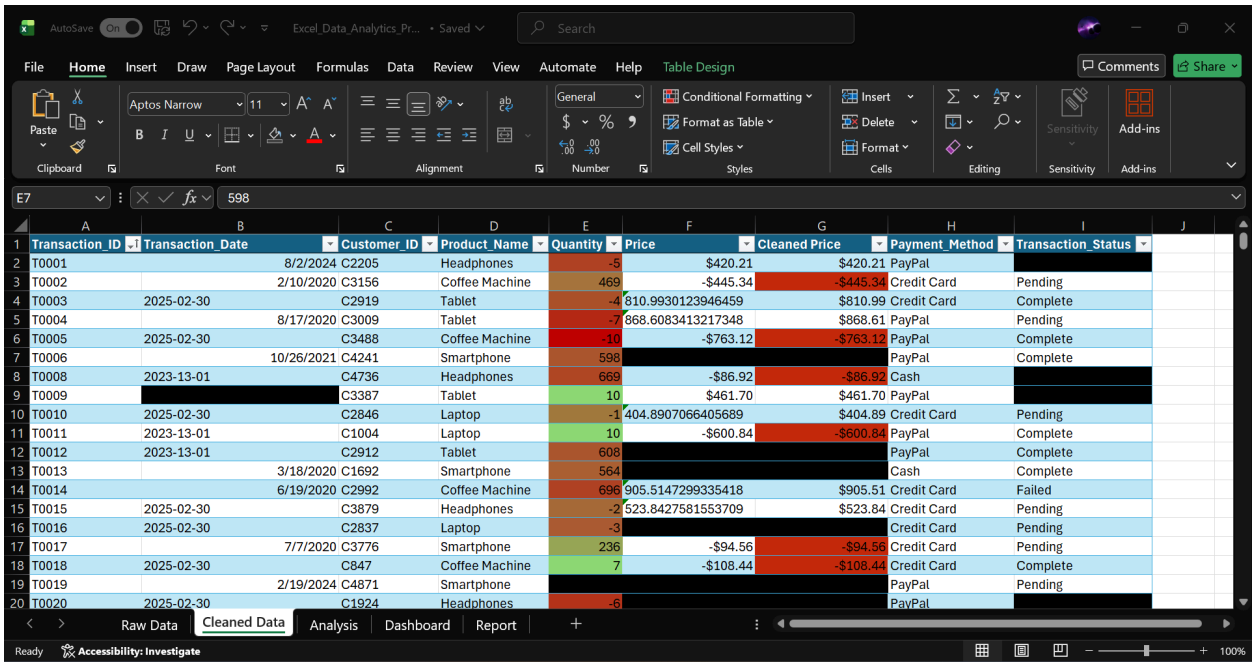
Before moving to Phase 3, ask yourself:

1. Have I addressed every problem I identified?
2. Did I accidentally change information that shouldn't have been changed?
3. Are the values consistent?
4. Are the data types correct?
5. Are the dates recognized correctly?
6. Are unnecessary columns removed?
7. Can the dataset now be analyzed without obvious data-quality problems?

If yes, your dataset is **analysis-ready**.

Cleaning Assessment for the Dirty Transactions Financial Worksheet

Ending Example of Cleaned Data Set



	A	B	C	D	E	F	G	H	I	J
1	Transaction_ID	Transaction_Date	Customer_ID	Product_Name	Quantity	Price	Cleaned Price	Payment_Method	Transaction_Status	
2	T0001	8/2/2024	C2205	Headphones	-5	$420.21	$420.21	PayPal		
3	T0002	2/10/2020	C3156	Coffee Machine	469	-$445.34	-$445.34	Credit Card	Pending	
4	T0003	2025-02-30	C2919	Tablet	-4	810.9930123946459	$810.99	Credit Card	Complete	
5	T0004	8/17/2020	C3009	Tablet	-7	868.6083413217348	$868.61	PayPal	Pending	
6	T0005	2025-02-30	C3488	Coffee Machine	-10	-$763.12	-$763.12	PayPal	Complete	
7	T0006	10/26/2021	C4241	Smartphone	598			PayPal	Complete	
8	T0008	2023-13-01	C4736	Headphones	669	-$86.92	-$86.92	Cash		
9	T0009		C3387	Tablet	10	$461.70	$461.70	PayPal		
10	T0010	2025-02-30	C2846	Laptop	-1	404.8907066405689	$404.89	Credit Card	Pending	
11	T0011	2023-13-01	C1004	Laptop	10	-$600.84	-$600.84	PayPal	Complete	
12	T0012	2023-13-01	C2912	Tablet	608			PayPal	Complete	
13	T0013	3/18/2020	C1692	Smartphone	564			Cash	Complete	
14	T0014	6/19/2020	C2992	Coffee Machine	696	905.5147299335418	$905.51	Credit Card	Failed	
15	T0015	2025-02-30	C3879	Headphones	-2	523.8427581553709	$523.84	Credit Card	Pending	
16	T0016	2025-02-30	C2837	Laptop	-3			Credit Card	Pending	
17	T0017	7/7/2020	C3776	Smartphone	236	-$94.56	-$94.56	Credit Card	Pending	
18	T0018	2025-02-30	C847	Coffee Machine	7	-$108.44	-$108.44	Credit Card	Complete	
19	T0019	2/19/2024	C4871	Smartphone				PayPal	Pending	
20	T0020	2025-02-30	C1924	Headphones	-6			PayPal		

Phase 2 Workflow

Your overall process should now be:

Raw Dataset

↓

Review Kaggle Documentation

↓

Inspect Dataset Yourself

↓

Identify Actual Data-Quality Issues

↓

Create Cleaning Plan

↓

Copy Raw Data → Cleaned Data

↓

Clean Only the Issues That Apply

↓

Perform Final Quality Check

↓

Analysis-Ready Dataset

Phase 3 — Formulas & Analytical Calculations

Using Excel Formulas to Answer Business Questions

In Phase 1, you were introduced to basic Excel formulas and functions. Now you will build on those fundamentals and use formulas to **analyze the cleaned dataset**.

The goal of this phase is not to memorize a list of formulas.

Instead, you will learn to look at a question such as:

> **"What is the total revenue generated by each region?"**

and determine:

1. What information the question is asking for.
2. Which column(s) contain that information.
3. Which Excel function can answer the question.
4. How to write the formula.
5. How to interpret the result.

Important

Do not simply copy the examples below.

Use your own dataset's columns and values. If your dataset doesn't contain a particular type of information, adapt the question to something your dataset can actually answer.

3.1 Before You Start: Identify Your Dataset Columns

Before practicing formulas, review your **Cleaned Data** sheet.

Create a quick reference table containing the columns you will use most often.

Column	What It Represents	Data Type
_____	_____	Text
_____	_____	Number
_____	_____	Date
_____	_____	Category
_____	_____	Currency

For example:

Column	What It Represents	Data Type
Sales	Revenue from transaction	Currency
Region	Customer region	Text
Product	Product purchased	Text
Quantity	Units purchased	Number
Order Date	Date of transaction	Date

This will make the rest of Phase 3 easier because you'll know what information each column contains.

3.2 Set Up an Analysis Area

Don't perform your analysis directly inside the raw dataset unless you specifically need a helper column.

Go to the **Analysis** sheet you created earlier.

At the top, create:

Formula Analysis

Then organize your work into sections.

For example:

Question	Formula Used	Result
What is the total revenue?	SUM	_____
What is the average revenue?	AVERAGE	_____
How many transactions are there?	COUNT	_____

This allows your Analysis sheet to eventually become the foundation for your dashboard and business report.

3.3 SUM

What does it do?

`SUM` adds numerical values together.

You already encountered this in Phase 1. Now you'll use it to answer an actual analytical question.

Syntax

=SUM(range)

Example

If revenue is in column E:

=SUM(E2:E100)

Dataset Question

> **What is the total revenue/sales generated by the entire dataset?**

Steps

1. Go to the **Analysis** sheet.
2. Create a label:

   Total Revenue

3. Click the cell next to it.
4. Enter your `SUM` formula using the appropriate column.
5. Press **Enter**.
6. Format the result as currency if appropriate.
7. Label the result clearly.

What should you learn?

You should be able to look at:

> "What is the total?"

and immediately recognize:

> **SUM**

3.4 AVERAGE

What does it do?

`AVERAGE` calculates the arithmetic mean of a group of numbers.

Syntax

=AVERAGE(range)

Dataset Question

> **What is the average _____ per transaction/customer/order?**

Choose a numerical column from your dataset.

For example:

=AVERAGE(E2:E100)

Steps

1. Identify the numerical column you want to analyze.
2. Create a label on the Analysis sheet.
3. Enter the `AVERAGE` formula.
4. Press **Enter**.
5. Format the result appropriately.
6. Write one sentence explaining what the number means.

Interpretation

Don't stop at:

> Average revenue = $425.32

Write:

> **The average transaction generated approximately $425.32 in revenue.**

This is an important analytical habit: **a number without context isn't an insight.**

3.5 COUNT

What does it do?

`COUNT` counts cells containing **numbers**.

Syntax

=COUNT(range)

Dataset Question

> **How many numerical records are in this column?**

For example:

=COUNT(E2:E100)

This is useful for understanding how many observations contain numerical data.

Important

`COUNT` counts **numbers**, not text.

For example:

Value
100
200
Apple
300

=COUNT(A1:A4)

would return:

33

because only three cells contain numbers.

3.6 COUNTA

What does it do?

`COUNTA` counts cells that **aren't empty**.

That means it can count:

- Text
- Numbers
- Dates
- Other non-empty values

Syntax

=COUNTA(range)

Dataset Question

> **How many records contain a value in this column?**

For example, if every customer should have a name:

=COUNTA(A2:A100)

You could use this to help determine whether records are missing information.

COUNT vs. COUNTA

This distinction is important:

COUNT

> How many cells contain numbers?

COUNTA

> How many cells contain anything?

3.7 MIN and MAX

These functions help identify the smallest and largest values in a dataset.

MIN

=MIN(range)

MAX

=MAX(range)

Dataset Questions

> **What is the smallest _____ in the dataset?**

and

> **What is the largest _____ in the dataset?**

For example:

=MIN(E2:E100)

and

=MAX(E2:E100)

Analytical Application

If you're analyzing sales:

MIN → smallest transaction

MAX → largest transaction

If you're analyzing wait times:

MIN → shortest wait

MAX → longest wait

If you're analyzing inventory:

MIN → lowest inventory level

MAX → highest inventory level

3.8 IF

Now we're moving beyond basic calculations.

What does IF do?

`IF` allows Excel to make a decision.

It basically says:

> **IF this condition is true, do this. Otherwise, do something else.**

Syntax

=IF(condition, value_if_true, value_if_false)

Example

Suppose you want to determine whether a transaction is considered "High Value."

=IF(E2>=500,"High Value","Standard")

Excel checks:

> Is E2 greater than or equal to $500?

If **yes**:

> High Value

If **no**:

> Standard

Dataset Question

Create a useful category based on one of your numerical columns.

For example:

> **Which transactions should be classified as high-value?**

Steps

1. Add a new column to the right of your dataset.
2. Give it a descriptive name such as **Transaction Category**.
3. Write your IF formula.
4. Press Enter.
5. Fill the formula down.
6. Review the results.

3.9 IFS

`IFS` is useful when you have **multiple conditions**.

Instead of:

> IF this, otherwise that

you can create multiple categories.

Syntax

=IFS(

condition1,result1,

condition2,result2,

condition3,result3

)

Example

Suppose you want to classify revenue:

=IFS(

E2>=1000,"High",

E2>=500,"Medium",

E2<500,"Low"

)

This creates:

Revenue	Category
$1,250	High
$750	Medium
$300	Low

Dataset Question

> **How can I divide my records into meaningful categories based on a numerical value?**

Choose thresholds that make sense for **your dataset**.

Important

Don't randomly choose categories just to practice the function. Think about what would actually be useful in an analysis.

3.10 AND / OR

These functions allow you to evaluate **multiple conditions**.

AND

`AND` requires **all conditions** to be true.

For example:

=AND(E2>=500,F2="North")

This asks:

> Is the revenue at least $500 **AND** is the region North?

Both must be true.

OR

`OR` requires **at least one condition** to be true.

=OR(F2="North",F2="South")

This asks:

> Is the region North **OR** South?

Only one needs to be true.

Using AND/OR with IF

This is where they become particularly useful.

Example

=IF(AND(E2>=500,F2="North"),"Priority","Standard")

Now you're telling Excel:

> If the transaction is at least $500 **and** comes from the North region, classify it as Priority.

Dataset Question

> **Can I identify records that meet multiple business conditions?**

Create a useful classification based on your dataset.

3.11 SUMIF

Now we're getting into **conditional analysis**.

What does SUMIF do?

`SUMIF` adds values **only when they meet a condition**.

Syntax

=SUMIF(criteria_range,criteria,sum_range)

Example

Suppose:

- Region = Column B
- Revenue = Column E

To calculate revenue from the North:

=SUMIF(B:B,"North",E:E)

Excel asks:

> Which rows have Region = North?

Then:

> Add the corresponding revenue.

Dataset Question

> **What is the total revenue for each region?**

Create a small table:

Region	Total Revenue
North	_____

South _____

East _____

West _____

Then use `SUMIF` to calculate each value.

3.12 SUMIFS

`SUMIFS` is similar to `SUMIF`, but it allows you to use **multiple conditions**.

Example

> What is the total revenue from the North region for Product A?

=SUMIFS(E:E,B:B,"North",C:C,"Product A")

Excel is essentially doing:

> Find rows where Region = North
> **AND** Product = Product A
> → Add their revenue.

Dataset Question

Create your own question requiring **at least two conditions**.

For example:

> **What is the total _____ for [category] in [region]?**

3.13 COUNTIF

`COUNTIF` counts how many records meet **one condition**.

Syntax

=COUNTIF(range,criteria)

Example

=COUNTIF(B:B,"North")

This answers:

> **How many records are from the North region?**

Dataset Question

> **How many records belong to each category/region/product?**

Create a table:

Region	Number of Records
North	_____
South	_____
East	_____
West	_____

3.14 COUNTIFS

`COUNTIFS` counts records that meet **multiple conditions**.

Example

=COUNTIFS(B:B,"North",C:C,"Product A")

This asks:

> How many records are for Product A **AND** come from the North?

Dataset Question

Create a question involving two or more conditions.

For example:

> **How many customers from the North purchased Product A?**

or:

> **How many transactions exceeded $500 in the North region?**

3.15 XLOOKUP

Now we're moving into formulas used to **retrieve information**.

What does XLOOKUP do?

XLOOKUP searches for something and returns related information from another column.

Think:

> **"I know this value. Find it and tell me something else about that record."**

Example

Suppose you have:

Product ID	Product	Price
P001	Laptop	$900
P002	Monitor	$300

P003 Keyboard $75

If you know the Product ID and want the product name:

=XLOOKUP(A2,A:A,B:B)

It searches for the value in A2 in column A and returns the corresponding value from column B.

Dataset Question

Create a question where you need to **retrieve information based on an identifier**.

For example:

> **Given a Customer ID, what is the customer's region?**

or:

> **Given a Product ID, what is the product name?**

Steps

1. Identify the lookup value.
2. Identify the column where Excel should search.
3. Identify the column containing the information you want returned.
4. Write the XLOOKUP formula.
5. Test it with several records.

3.16 INDEX/MATCH

`INDEX/MATCH` is another way to retrieve information.

You may encounter this combination in older Excel workbooks and professional environments, so it is useful to understand even if you prefer XLOOKUP.

MATCH

`MATCH` finds the **position** of a value.

=MATCH(A2,B:B,0)

INDEX

`INDEX` returns a value from a specific position.

=INDEX(C:C,5)

Together:

=INDEX(C:C,MATCH(A2,B:B,0))

This means:

> Find A2 in column B, determine where it is, and return the corresponding value from column C.

Why learn both?

XLOOKUP is generally easier to read and use.

But understanding `INDEX/MATCH` helps you:

- Work with older Excel files
- Understand lookup logic
- Troubleshoot existing spreadsheets
- Recognize formulas used in professional workbooks

3.17 Text Functions

Text functions help clean, manipulate, and analyze text.

You already encountered `TRIM` during Phase 2. Now you'll use text functions analytically.

Functions to Practice

LEFT

Returns characters from the beginning of text.

=LEFT(A2,3)

RIGHT

Returns characters from the end.

=RIGHT(A2,3)

MID

Returns characters from the middle.

=MID(A2,2,4)

LEN

Counts the number of characters.

=LEN(A2)

UPPER

=UPPER(A2)

LOWER

=LOWER(A2)

PROPER

=PROPER(A2)

TRIM

=TRIM(A2)

Dataset Challenge

Choose a text column and answer a question such as:

> **How many characters are in each customer name?**

or:

> **Can I extract the first three characters of each product code?**

or:

> **Can I standardize the capitalization of this category?**

3.18 Date Functions

Dates can provide valuable information for analysis.

Some useful functions include:

YEAR

Extracts the year:

=YEAR(A2)

MONTH

Extracts the month:

=MONTH(A2)

DAY

Extracts the day:

=DAY(A2)

TODAY

Returns today's date:

=TODAY()

DAYS

Calculates the number of days between two dates:

=DAYS(B2,A2)

Dataset Challenge

Use the date column in your dataset to answer a question such as:

> **How many years/months of data are represented in the dataset?**

or:

> **Which month had the highest number of transactions?**

or:

> **How many days passed between two events?**

If your dataset contains transaction dates, you can create helper columns for:

- Year
- Month
- Day
- Day of week

These can become especially useful when you reach **Pivot Tables and Dashboard Creation**.

3.19 Formula Challenge — Solve the Question First

This is the most important part of Phase 3.

Now that you've practiced the formulas, **stop starting with the formula.**

Start with the **business question**.

For each question:

Step 1 — Read the question

Example:

> **What is the total revenue generated by each region?**

Step 2 — Identify what you need

You need:

- A **Region** column
- A **Revenue** column

Step 3 — Determine the type of calculation

You're adding revenue based on a condition.

Step 4 — Choose the function

That's:

SUMIF\boxed{\text{SUMIF}}

Step 5 — Build the formula

=SUMIF(RegionColumn,"North",RevenueColumn)

Step 6 — Interpret the answer

Don't just write:

> $52,430

Write:

> **The North region generated $52,430 in total revenue.**

3.20 Your Formula Practice Questions

Now create a set of questions using your **actual dataset**.

Try to answer at least one question with each category of formula.

Formula	Question to Solve
SUM	What is the total _____?
AVERAGE	What is the average _____?
COUNT	How many numerical records are there?

COUNTA	How many non-empty records are there?
MIN	What is the smallest _____?
MAX	What is the largest _____?
IF	How can I classify records based on one condition?
IFS	How can I classify records into multiple groups?
AND	Which records meet condition A **and** condition B?
OR	Which records meet condition A **or** condition B?
SUMIF	What is the total _____ for each category?
SUMIFS	What is the total _____ when multiple conditions are met?
COUNTIF	How many records meet this condition?
COUNTIFS	How many records meet multiple conditions?
XLOOKUP	What information corresponds to this ID/value?
INDEX/MATCH	Can I retrieve the same information using another lookup method?
Text Functions	What information can I extract or standardize from text?
Date Functions	What patterns can I identify from the dates?

Phase 3 Complete

By the end of this phase, you should be able to look at a dataset and think:

> **"What question am I trying to answer?"**

rather than:

> **"Which formula am I supposed to use?"**

That's the actual analytical skill you're trying to build.

Your workflow is now:

Raw Data
↓
Data Cleaning
↓
Cleaned Data
↓
Ask a Question
↓
Choose the Appropriate Formula
↓
Calculate
↓
Interpret the Result
↓

Business Insight

And this sets you up perfectly for **Phase 4 — Pivot Tables & Analysis**, where instead of manually writing formulas for every category, you'll learn how Excel can summarize much larger amounts of data efficiently.

Phase 4 — Pivot Tables

Turning a Clean Dataset into Meaningful Insights

In the previous phases, you cleaned your dataset and learned how to use formulas to answer individual questions.

Now you'll learn how to use **PivotTables** to summarize large amounts of data quickly.

A PivotTable allows you to take hundreds or thousands of rows and reorganize them into a summary that makes patterns easier to see.

For example, instead of manually calculating revenue for every region with `SUMIF`, you can create a PivotTable that automatically summarizes:

Region	Total Revenue
East	$125,400
North	$143,200
South	$98,700
West	$167,800

You can then change the PivotTable to analyze the same dataset by:

- Product
- Month
- Customer
- Category
- Year
- Region
- Average
- Count
- Sum

The goal of this phase is to move from:

> **"How do I calculate this?"**

to:

> **"What does this data tell me?"**

4.1 — Understand the Purpose of PivotTables

A PivotTable summarizes large amounts of data so you can identify patterns without writing a separate formula for every question.

For this project, PivotTables will help you investigate:

- Which products have the highest average quantities?
- How are transactions distributed across statuses?
- Which products have the most suspicious quantity records?

Goal: Learn to summarize and interpret your existing dataset.

4.2 — Understand the Four PivotTable Areas

When you create a PivotTable, Excel provides four areas:

- **Rows:** The categories you want to group together.
- **Columns:** Additional categories you want to compare side by side.
- **Values:** The numbers Excel calculates, such as sums, averages, and counts.
- **Filters:** Options for limiting the records included in the analysis.

You do not need to use all four areas in every PivotTable.

4.3 — Create Your First PivotTable

1. Open your **Cleaned Data** worksheet.
2. Click any cell inside your dataset.
3. Select **Insert** → **PivotTable**.
4. Verify that Excel selected the correct dataset.
5. Select **New Worksheet**.
6. Click **OK**.
7. Rename the new worksheet **Pivot Analysis**.

You can place all three PivotTables on this worksheet, leaving enough space between them.

4.4 — PivotTable #1: Quantity by Product

Business question: How do quantities vary across the different products?

Set up the PivotTable

- **Rows:** Product Name
- **Values:** Quantity → Average
- **Values:** Transaction ID → Count

Make sure the Quantity field uses **Average**, not Sum. Make sure Transaction ID uses **Count**.

Analyze your results

1. Which product has the highest average quantity?
2. Which product has the lowest average quantity?
3. How many transactions are associated with each product?
4. Do some products have substantially higher average quantities than others?

Important: A high average quantity does not automatically mean something is wrong. Use your findings to identify products that may deserve further investigation.

4.5 — PivotTable #2: Transactions by Status

Business question: How are transactions distributed across transaction statuses?

Set up the PivotTable

- **Rows:** Transaction_Status
- **Values:** Transaction ID → Count

If you have blank statuses, leave them visible so you can identify missing information.

Analyze your results

1. How many transactions fall under each status?
2. How many transactions have a missing status?
3. Which status appears most frequently?
4. Does the number of missing statuses suggest that additional investigation is needed?

This PivotTable helps you summarize transaction records without assuming that a particular status is incorrect.

4.6 — PivotTable #3: Suspicious Quantities by Product

Business question: Which products have the most transactions flagged for suspicious quantities?

First, make sure your dataset contains a helper column that classifies quantities as **Suspicious** or **Normal**.

For example, if Quantity is in column E:

```
=IF(OR(E2<0,E2>300),"Suspicious","Normal")
```

The threshold of 300 is a preliminary screening rule for this project, not proof that a transaction is erroneous.

Set up the PivotTable

- **Rows:** Product Name
- **Columns:** Quantity Flag
- **Values:** Transaction ID → Count

Your PivotTable should show the number of normal and suspicious transactions for each product.

Analyze your results

1. Which product has the most suspicious quantity records?
2. Which products have no suspicious quantities?
3. Are suspicious records concentrated in particular products?
4. Which records should be examined individually?

Important: A PivotTable identifies patterns; it does not establish why a record is suspicious. Negative quantities, for example, could represent returns or reversals.

4.7 — Format Your PivotTables

Make your PivotTables easy to read.

- Give each PivotTable a descriptive title.
- Adjust column widths.
- Apply a readable PivotTable style.
- Format numerical values consistently.
- Rename value fields where appropriate.
- Keep blank statuses visible when analyzing missing information.

Avoid adding unnecessary formatting that makes the results harder to interpret.

4.8 — Use Filters to Investigate Records

Filters let you focus on specific parts of your dataset.

For example, you can filter the quantity PivotTable to examine one product or use the quantity flag to focus on suspicious records.

Practice selecting a filter, observing the results, and then returning the PivotTable to show all records.

You do not need to build a separate PivotTable for every filter you try.

4.9 — Refresh Your PivotTables

If you change the underlying data, refresh your PivotTables so that they reflect those changes.

1. Click inside the PivotTable.
2. Right-click.
3. Select **Refresh**.

If you add new records beyond the original dataset range, verify that the PivotTable's source range includes those records before refreshing.

4.10 — Summarize Your Findings

Create a small summary table containing your three analyses and their most important findings.

Analysis	Key Finding
Quantity by Product	Which product has the highest average quantity?
Transactions by Status	Which status is most common, and how many statuses are missing?
Suspicious Quantities by Product	Which product has the most records flagged for investigation?

Write one or two sentences about each PivotTable using your actual results.

Distinguish between what the data shows and what you suspect might be happening. For example, a product with many suspicious quantities deserves investigation, but that does not establish that the records are incorrect.

Phase 4 Completion Checklist

- I understand Rows, Columns, Values, and Filters.
- I created three PivotTables using my actual dataset fields.
- I can summarize data using Average and Count.
- I can filter and refresh a PivotTable.
- I formatted my PivotTables for readability.
- I wrote findings based on my actual results.

Phase 5 — Dashboard Creation

Turning Your Analysis into a Visual Summary

5.1 — Understand the Purpose of a Dashboard

A dashboard presents the most important findings from your dataset in one place.

For this project, your dashboard should help someone quickly understand:

- How many transaction records are in the dataset?
- How many records have suspicious quantities?
- How many transactions have missing statuses?
- Which products may require further investigation?

Goal: Present your findings clearly without making unsupported claims about the data.

5.2 — Define Your Dashboard's Purpose

Your hypothetical stakeholder is a business manager who wants to understand potential problems in transaction data.

Use this objective:

> This dashboard was created to summarize transaction data, highlight potential data-quality issues, and identify records that may require further investigation.

Your dashboard is an **investigation tool**, not proof that the flagged transactions are errors.

5.3 — Decide What Your Dashboard Needs

Keep the dashboard simple. It needs only two main components:

1. **Summary metrics:** Important numbers that provide a quick overview.
2. **Charts:** Visuals that help you compare products and identify patterns.

You do not need multiple filters, a date timeline, or a large collection of charts.

5.4 — Create Your Dashboard Worksheet

1. Click the **+** button to create a new worksheet.
2. Rename the worksheet `Dashboard`.

3. Move it to the beginning of your workbook if desired.
4. Add the title:
   Transaction Data Quality Dashboard
5. Add a short subtitle:
   Summary of transaction records and potential data-quality issues.

This worksheet will become the main page for your project.

5.5 — Add Your Summary Metrics

Start with three or four summary metrics.

Metric	What It Tells You
Total Transaction Records	How many transaction records are in the dataset?
Suspicious Quantity Records	How many records meet your suspicious-quantity rule?
Negative Price Records	How many records contain negative prices?
Missing Transaction Statuses	How many records have no transaction status?

Use the results from your existing formulas or PivotTables rather than manually typing numbers.

Important: A negative price may represent a credit or reversal, and a suspicious quantity may have a legitimate explanation. Label these metrics as potential issues, not confirmed errors.

If you cannot verify a metric or its calculation, leave it out until it is ready.

5.6 — Create a Chart of Suspicious Quantities by Product

Use the PivotTable you created in Phase 4.

Purpose: Identify which products have the most records flagged for suspicious quantities.

1. Click inside your suspicious-quantity PivotTable.
2. Select **Insert → PivotChart**.
3. Choose a bar or column chart.
4. Give the chart the title **Suspicious Quantity Records by Product**.
5. Make sure the chart displays the count of suspicious records, not the sum of their quantities.

This chart will help you identify which products deserve closer examination.

5.7 — Create a Chart of Transaction Statuses

Use your transaction-status PivotTable.

Purpose: Show how transaction records are distributed across statuses, including missing statuses if present.

1. Click inside the transaction-status PivotTable.
2. Select **Insert** → **PivotChart**.
3. Choose a bar or column chart.
4. Give it the title **Transaction Records by Status**.
5. Make sure blank statuses remain visible if your PivotTable includes them.

This chart can reveal whether missing status information is a noticeable issue in your dataset.

If the chart contains too many categories or is difficult to read, keep the PivotTable instead. A chart is not necessary when it doesn't improve understanding.

5.8 — Arrange and Format Your Dashboard

Arrange the dashboard so the most important information is easy to find.

Suggested layout:

- **Top:** Dashboard title and subtitle.
- **Second row:** Three or four summary metric cards.
- **Middle:** Suspicious Quantity Records by Product chart.
- **Bottom:** Transaction Records by Status chart.

Keep the design consistent:

- Use readable text.
- Give charts descriptive titles.
- Align the metric cards and charts.
- Format numbers consistently.
- Avoid unnecessary colors and decorative elements.
- Hide gridlines by going to **View** → **Gridlines** and unchecking the option.

The goal is clarity, not elaborate graphic design.

5.9 — Write a Short Dashboard Summary

Write a short paragraph describing the most important findings.

Answer these questions using your actual results:

1. How many transaction records were analyzed?
2. How many records were flagged for suspicious quantities?
3. How many records had negative prices or missing statuses?
4. Which products had the most suspicious quantity records?
5. What should be investigated next?

Do not claim that a transaction is incorrect simply because it was flagged. Explain that the records require verification.

5.10 — Complete Your Dashboard

Use this checklist before finishing Phase 5.

- Created a Dashboard worksheet.
- Added a clear dashboard title.
- Defined the purpose of the dashboard.
- Added three or four verified summary metrics.
- Created a chart of suspicious quantities by product.
- Created a chart of transaction statuses, if useful.
- Formatted the dashboard for readability.
- Wrote a short summary of the findings.
- Checked that the displayed numbers match the underlying analysis.
- Clearly distinguished potential issues from confirmed errors.

Phase 5 Complete

Your project now follows this workflow:

Raw Data → Cleaned Data → Formulas → PivotTables → Dashboard → Findings

You do not need to add more features simply to make the project look advanced. A small, accurate dashboard that communicates your findings is better than a complicated dashboard built around metrics your dataset cannot support.

Phase 6 — Business Reporting

Turning Data Analysis into Business Recommendations

You've now cleaned your dataset, performed calculations, built PivotTables, and created a dashboard.

The final step is to communicate what you discovered and explain what someone should investigate next.

A stakeholder doesn't necessarily need to know every Excel function you used. They need to understand:

- What did you find?
- Why does it matter?
- What should happen next?

Your goal: Create a concise business report that summarizes your analysis of potential data-quality issues in the transaction dataset.

6.1 Understand the Difference Between Analysis and Reporting

Your analysis identifies what the data shows. Your report explains why those findings matter and what actions might be appropriate.

For example:

Analysis:

> Several transactions contain negative or unusually high quantities.

Interpretation:

> These records may indicate data-entry errors, returns, adjustments, or unusual transactions that require review.

Recommendation:

> Review the flagged records against the original transaction records and confirm whether the quantities are valid.

Notice that the recommendation doesn't automatically assume the records are incorrect.

Your task: Keep the distinction between facts, interpretations, and recommendations clear throughout your report.

6.2 Establish Your Business Question

Your report needs one overarching question that explains the purpose of your analysis.

Use this question:

> **What potential data-quality issues exist in the transaction dataset, how extensive are they, and what should be investigated next?**

This question fits the work you've completed because you've examined transaction records, cleaned data, created formulas, and identified potentially suspicious values.

Your analysis is intended to identify records for investigation—not establish that every unusual value is an error.

6.3 Review Your Analysis and Identify Key Findings

Review your work from Phases 3–5:

- Formula calculations
- Data-quality flags
- PivotTables
- Dashboard
- Dataset quality assessment

Choose **three to five important findings**. You do not need to repeat every calculation or number in your workbook.

Use this table to organize your findings:

Finding	Evidence	Why It Matters

Some transactions contain suspicious quantities.	Count of records flagged by your quantity rule.	These records may need verification.
Some transactions contain negative prices.	Count of records with negative prices.	These values may represent adjustments, credits, returns, or errors.
Some transactions have missing statuses.	Count of records with blank transaction statuses.	Missing statuses may make transaction outcomes harder to determine.
Certain products have more flagged quantity records than others.	Results from your suspicious-quantity PivotTable.	Those products may deserve closer review.

Important: These are potential findings based on the issues you've been investigating. Use your actual results to determine which findings belong in your final report. Do not claim an issue exists if your analysis did not find it.

6.4 Separate Facts From Interpretations

This is one of the most important skills in data analysis.

A fact

Something directly supported by your data.

Example:

> The analysis identified 12 records with quantities flagged by the screening rule.

An interpretation

What the finding might mean.

Example:

> These records may contain unusual transactions or data-quality problems.

A hypothesis

A possible explanation that has not been verified.

Example:

> Some negative quantities may represent product returns.

A recommendation

An action someone could take.

Example:

> Compare the flagged records with the original transaction documentation to determine whether the quantities are valid.

Keep these categories separate.

Remember: A flagged record is not automatically an incorrect record.

6.5 Identify Business Implications and Data Limitations

For each major finding, explain why it matters.

For example:

Finding: Some transactions have missing statuses.

Potential implication: It may be difficult to determine whether these transactions were completed, canceled, or otherwise processed.

Recommended next step: Review the original records or consult the person responsible for maintaining the transaction data.

Also identify what your dataset cannot tell you.

For this project, possible limitations include:

- The dataset may not explain why a particular quantity or price is unusual.
- Negative quantities and prices may have legitimate business explanations.
- Your screening threshold for suspicious quantities is preliminary and has not been verified against an official business rule.
- The dataset may not contain the documentation needed to confirm whether flagged records are errors.
- The meaning of the price column must be confirmed before interpreting its sum as revenue.

These limitations do not make your analysis useless. Recognizing them demonstrates that you understand the boundaries of your conclusions.

6.6 Create Recommendations

Create **three recommendations** based on your findings.

Your recommendations should be specific, realistic, and connected to evidence.

Use this framework:

Finding → Implication → Recommended Action

Recommendation 1: Review Suspicious Quantities

Finding: Some transaction records contain negative or unusually high quantities.

Potential implication: These records may require additional review before the dataset can be relied on for further analysis.

Recommended action: Compare flagged records with original transaction documentation and confirm the appropriate treatment of unusual quantities.

Recommendation 2: Investigate Negative Prices

Finding: Some records contain negative prices.

Potential implication: These values could affect price-related calculations if their meaning is not understood.

Recommended action: Determine whether negative prices represent legitimate credits, returns, adjustments, or data-entry errors before changing or excluding them.

Recommendation 3: Investigate Missing Transaction Statuses

Finding: Some transaction records may have missing statuses.

Potential implication: Missing information may make it harder to understand the outcome of those transactions.

Recommended action: Review the original records and determine whether the missing statuses can be verified and corrected.

Your task: Use your actual findings to decide which recommendations are appropriate. If your analysis did not identify one of these issues, replace that recommendation with one supported by your results.

6.7 Prioritize Your Recommendations

Not every issue needs the same level of attention.

Create a simple priority table:

Priority	Recommendation	Reason
High	Review the most concerning flagged records.	These records may affect the reliability of subsequent analysis.
Medium	Investigate missing transaction information.	Additional information may be needed to interpret certain records.
Low	Improve documentation of data-validation rules.	Clear rules may help identify unusual records consistently in the future.

These priorities are examples, not conclusions established by your dataset.

Adjust them based on your findings and the potential consequences of each issue.

You do not need to build a complicated risk-scoring system for this project.

6.8 Write the Final Business Report

Create a document titled:

Transaction Data Quality Analysis Report

Include the following sections.

1. Executive Summary

Briefly explain the purpose of your analysis, the most important findings, and your recommendations.

2. Business Question

State the question your analysis was designed to answer:

> What potential data-quality issues exist in the transaction dataset, how extensive are they, and what should be investigated next?

3. Key Findings

Summarize three to five findings and support each with actual results from your workbook.

4. Business Implications

Explain why the findings matter and how they could affect the interpretation or use of the transaction data.

5. Recommendations

List your three recommendations and briefly explain the evidence supporting each one.

6. Data Limitations

Explain what you could not verify from the available data.

7. Conclusion

Summarize the overall condition of the dataset and identify the most important next step.

You do not need to create a separate section for every formula, PivotTable, or chart. Refer to those results when they support an important finding.

6.9 Write Your Executive Summary

The executive summary is a short overview of the entire report. Write it after reviewing your results so that it accurately reflects your findings.

Use this template:

Executive Summary

This analysis examined transaction data to identify potential data-quality issues and determine which records may require further investigation. The analysis included data cleaning, formula-based checks, PivotTables, and a dashboard summarizing selected transaction patterns.

The analysis identified **[insert your most important findings]**. In particular, **[insert a finding supported by your results]**. These findings indicate that certain records may require additional verification before the dataset is used for further analysis.

Based on the available information, the recommended next steps are to **[recommendation 1]**, **[recommendation 2]**, and **[recommendation 3]**. However, the analysis cannot independently confirm that every flagged record is incorrect. Additional documentation or clarification of the relevant business rules may be necessary to determine the appropriate treatment of unusual records.

Replace the bracketed sections with your actual results. Do not insert numbers you have not verified.

6.10 Final Quality Check

Before completing Phase 6, make sure you have:

- Defined the overall business question.
- Reviewed the results from Phases 3–5.
- Identified three to five important findings.
- Supported findings with actual data.

- Distinguished facts from interpretations and hypotheses.
- Explained why the findings matter.
- Created three evidence-based recommendations.
- Identified important limitations in the dataset.
- Written an executive summary.
- Checked that the numbers in your report match your workbook.
- Avoided presenting potential errors as confirmed errors.

Phase 6 Complete

Your project now follows the full analytical workflow:

Raw Data → Data Cleaning → Formulas → PivotTables → Dashboard → Business Reporting → Recommendations

You have practiced more than Excel functions. You've learned how to investigate a dataset, communicate evidence, recognize uncertainty, and recommend appropriate next steps.

Your final deliverable: A completed Excel workbook containing your analysis and dashboard, accompanied by a concise business report explaining your findings and recommendations.